

Efficacies of vegetation litter and roots in strengthening rainfall infiltration for different stand ages on the Loess Plateau

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ABSTRACT

Vegetation litter cover and root systems have important effects on hillslope hydrological processes. Their relative importance may change with the stage of growth after afforestation. However, limited information is available on the dynamics of the quantitative contributions of litter and roots to rainfall infiltration. In this study, *Robinia pseudoacacia* plantations differing in stand age (18, 25, 30, and 40 years) and fallow slope land (control) were selected to investigate the influence of litter cover and roots on infiltration processes on hillslopes following simulated rainfall. Both the accumulation and decomposition rates of litter and the biomass of roots increased with stand age. Under the same simulated rainfall, infiltration in the plantations initially increased and thereafter decreased with stand growth, which was 2.93–4.91 times that of the fallow slope land. With increase in stand age from 18 to 40 years, litter contributed 60%, 11%, 82% and 91% to the increase in infiltration, and roots contributed 40%, 89%, 18% and 9%, respectively, to aforementioned four stands. The average contribution of litter to the increase in infiltration was 1.56 times greater than that of roots. The contribution of litter generally increased with stand age. Litter cover and the root system played a crucial role in strengthening uniform infiltration and preferential infiltration, respectively. In the presence of litter, the contribution of preferential flow to total infiltration was 67% of that in the absence of litter. These results contribute to an improved understanding of the effects of afforestation on hillslope hydrological processes and provide a theoretical reference for the management of plantations.

1. Introduction

Afforestation is an effective measure to control soil erosion, improve soil structure, prevent land degradation, and restore ecosystems (Jackson and Hobbs, 2009; Chen et al., 2015). Accordingly, larger-scale afforestation has been performed worldwide in recent decades (Fry et al., 2017; Suding et al., 2015). Wei et al. (2018) demonstrated that alterations in vegetation cover resulted in an average global runoff variation of $30.7\% \pm 22.5\%$. However, the impact of vegetation restoration on regional water resources is influenced by multiple factors, leading to significant disparities and controversies in existing research findings. The hydrological response mechanisms to forest changes remain incompletely understood (Zhou et al., 2015a; Zhang et al., 2017). Most studies suggest that regional runoff decreases under the influence of vegetation restoration, whereas some research has found an

increase in runoff in certain areas following afforestation (Feng et al., 2012; Sun et al., 2006). China currently has the largest artificial forest area in the world, the fractional vegetation cover of the Loess Plateau increased from 31.6% in 1999 to approximately 65% in 2017 (Li et al., 2017). The annual runoff in the Yellow River Basin has decreased from 39.2 billion m³ in the twentieth century to 25.5 billion m³ in the twenty-first century, with the minimum annual runoff being less than 14 billion m³ (Deng et al., 2012). However, whether streamflow reduction will be alleviated or intensified with the progression of growth after afforestation is unknown. To sustainably utilize the limited water resources and protect the eco-environment, it is important to focus on how plantation stage influences hillslope hydrological processes (Harden and Scruggs, 2003; Neris et al., 2012).

Infiltration is a pivotal hydrological process in ecosystems, exerting a direct influence on the generation and magnitude of surface runoff, as

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well as on soil erosion and soil moisture content. Alterations in soil moisture content, in turn, affect the growth of vegetation (Sergio et al., 2019). Therefore, research on the relationship between vegetation and runoff can be initiated from the perspective of infiltration. Vegetation restoration significantly changes rainfall infiltration on hillslopes, a process in which litter cover and root systems play important roles (Freschet et al., 2013). Infiltration is primarily influenced by factors such as soil moisture content and soil hydraulic conductivity, with soil hydraulic conductivity being closely associated with soil bulk density and porosity (Smith and Parlange, 1978; Franzluebbers, 2002). Litter can change the soil porosity and bulk density through decomposition and can also impede the formation of surface crust during rainfall events, thereby increasing the soil hydraulic conductivity to enhance infiltration (Xia et al., 2019; Zeng et al., 2018). Roots are interspersed in compact soil and form biological pores, which increases the number of large soil pores and promotes infiltration (Bengough et al., 2006; Vannoppen et al., 2017). Both litter and roots affect infiltration by directly or indirectly changing soil characteristics. Litter mainly affects the surface soil, whereas roots mainly affect the soil layer in which they are distributed, which is determined by vegetation type. Apart from the aforementioned influence on infiltration by altering soil structure, litter can promote infiltration on hillslopes by retaining runoff (Zhu et al., 2020a). Many studies have shown that vegetation litter and roots significantly increase the volume of infiltration, but there is variation in the degree of effect (Wang et al., 2023). Seiwa et al. (2021) demonstrated that in secondary broadleaf plantations, the correlation between litter and infiltration rate as well as porosity was higher than that of root density. However, Zhu et al. (2020b) contradicted this finding in shrublands, suggesting that roots had a greater impact on infiltration performance. Both studies compared the effects of litter and roots by establishing correlations with infiltration-related parameters, and significant differences observed in the results across different vegetation types. The influence of litter and roots on infiltration in plantations may differ from the scenarios mentioned above. Therefore, the present study aimed to quantitatively distinguish the contributions of litter and roots to infiltration in plantations, providing a more direct comparison of their respective impacts on infiltration.

The accumulation and degree of decomposition of litter, as well as the distribution and direction of root system extension, change with increase in plantation age (Jia et al., 2017). Both litter accumulation and decomposition are positively correlated with forest age, which has been ranked as mature forest > near-mature forest > middle-aged forest > young forest (Li et al., 2023; Brantley et al., 2019). The production rate of fine roots is negatively correlated with stand growth, whereas less information is available on the changes in coarse roots with stand age (Andersen et al., 2008). Therefore, their impacts on infiltration might also change with the growth stage after afforestation, resulting in differences in infiltration rate among stands of different ages. Zema et al. (2021) showed that water infiltration was higher in older stands and lower in younger pine plantations. Different plantations may have different trends in infiltration with stand growth, and different thresholds may exist, warranting further investigation. Additionally, trends in the contribution rates of litter and roots to infiltration may also vary with stand growth. Therefore, guided by previous research, we monitored the trends in infiltration rates with stand growth and the spatio-temporal dynamics of the contributions of litter and roots to infiltration, and examined the impact of soil structural evolution on infiltration after afforestation.

Robinia pseudoacacia has been widely planted on the Loess Plateau since the 1950s owing to its favorable soil and water conservation properties. However, in the early stages of planting, little attention was paid to the effects on soil moisture and runoff. It has been demonstrated that the widespread planting of *R. pseudoacacia* may result in a reduction in runoff, and some semi-arid and arid regions experiencing issues such as soil desiccation and low tree survival rates (Mei et al., 2018). The controversy over whether the planting of *R. pseudoacacia* is beneficial for

the rational development and utilization of water and soil resources on the Loess Plateau, as well as how to manage it in the future, still persists. Therefore, *R. pseudoacacia* was selected to investigate the effect of stand age on hillslope hydrological processes. The aims of this study were (1) to investigate the evolution of litter, root, and soil systems with stand age, (2) to distinguish the quantitative contribution of litter and roots to the increase in infiltration, and (3) to evaluate the mechanism by which litter and roots influence rainfall infiltration. The results provide a basis for sustainable ecological management and vegetation establishment on the Loess Plateau.

2. Materials and methods

2.1. Study area

This study was conducted in the Caijiachuan watershed on the Loess Plateau in China, which is located in the middle reaches of the Yellow River (110°39'45"–110°47'5"E, 36°14'27"–36°18'23"N). The study area has an elevation of 900–1513 m and is part of the National Ecosystem Research Network of China. The experimental area is located in a semiarid continental climate zone, with mean annual precipitation of approximately 500 mm in 1985–2020, potential evapotranspiration of 1729 mm, average annual temperature of 10.2°C, and sunshine hours of approximately 2563 h (Mei et al., 2018). In 1991 and 2001, the region was subject to two large-scale afforestation events. The main land-use types in the region are forest, farmland, grassland, and orchards, and the main plantation species are *Robinia pseudoacacia*, *Pinus tabulaeformis*, and *Platycladus orientalis*.

2.2. Experimental design and data measurement

2.2.1. Simulated rainfall device

The simulated rainfall experiment utilized needle-type artificial rainfall devices with dimensions of 1.2 m × 1.2 m × 1.5 m (Fig. 1). The needles had a diameter of 0.55 mm, with evenly distributed pinholes. The rainfall water used was a bright blue solution with a concentration of 4 g/L. Throughout the simulated rainfall event, the water level in the rainfall device remained constant through water circulation, with a calibrated rainfall intensity of 30 mm/h and a rainfall uniformity coefficient exceeding 0.80. In comparison with tracer tests involving direct pouring or colorant spraying, the advantage of the present method was its ability to provide results that more accurately reflect the natural conditions and runoff outcomes of the slope.

2.2.2. Experimental design

For a field experiment with artificially simulated rainfall, five typical slopes with similar topographic characteristics were selected in four *R. pseudoacacia* plantations aged 18, 25, 30, and 40 years, and a fallow slope land (control), respectively. The main plants of each *R. pseudoacacia* sample plot were *Robinia pseudoacacia* and *Festuca arundinacea*. In contrast, the fallow slope land exhibited minimal vegetative cover. The slope gradient of each plot was about 15°, the aspect was southeast, and the soil type was loamy soil.

The four *R. pseudoacacia* plantations of different ages were sampled to assess litter retention and removal. Initially, two sets of littered and two sets of un-littered runoff plots were established in each *R. pseudoacacia* plantation, and two sets of un-littered runoff plots were established in fallow slope land. If significant differences were observed between the results of these replicate experiments, additional experiments were performed as a third set. The analysis focused on two sets of experiments with similar results, and the average of the results from the two replicate experiments was calculated. The runoff plots were 1 m × 1 m in size, with an approximate slope of 15°. The runoff plots were bordered by asbestos tiles, which were buried 20 cm into the soil. Fine soil was used to fill on both sides of the tiles to prevent water from infiltrating the inner wall of the tiles. Runoff collection devices were

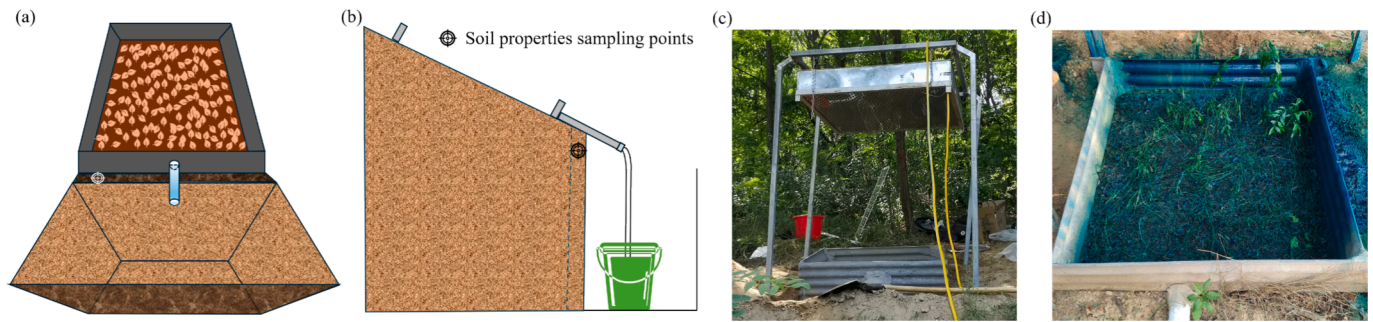


Fig. 1. Design of the experimental rainfall device. (a) Schematic diagram of runoff plot; (b) Side view of runoff plot; (c) Needle tube rainmaker; (d) Underlying surface condition.

positioned at the surface outlets of the runoff plots. Prior to the simulated rainfall, the underlying surface was carefully leveled to achieve a uniform and essentially flat profile, preventing water accumulation in local depressions and uneven infiltration resulting from uneven terrain (Fig. 1).

The artificial rainfall experiments were performed in July–September 2021 and 2022 under continuous sunny weather. Given the considerable depth of the soil layers in the Loess Plateau, infiltration excess runoff dominates, with the saturated hydraulic conductivity of the surface soil typically exceeding 25–30 mm/h (Zhu et al., 2022; Wang et al., 2013). And considering the need for both infiltration and runoff processes in different terrains in comparative experiments, a rainfall intensity of 30 mm/h was selected. Each rainfall event lasted 3 h, and the rainfall volume was 90 mm. The interval between rainfall events was approximately 24 h, and three consecutive rainfall events were completed within 72 h, for total rainfall of 270 mm. Given the loss of seams and needles during transportation, installation, and use of the rain gauge, the total rainfall was calculated to be reduced by approximately 20%, so the actual total rainfall was considered to be 216 mm. During the rainfall event, the surface runoff was collected with collecting buckets, and the volumes of each was measured with measuring cylinders. At the end of each daily rainfall event, the surface and profile of the plots were covered with waterproof canvas to reduce the impact of water evaporation. At 24 h after the completion of three rainfall events in each experimental plot, a vertical section of 1 m × 1 m was excavated in the middle of the slope. A non-deformable aluminum alloy frame was then placed around the profile for geometric correction and standardization of later image processing. The was photographed in a shaded site.

2.2.3. Data collection

Prior to commencing the simulated rainfall experiment, we investigated the distribution characteristics of litter and soil structure in the selected plots. For the litter cover survey, the thickness of the non-decomposed and semi-decomposed layers, as well as the litter density, were measured. A 10 m × 10 m plot was selected in each *R. pseudoacacia* plantation, where three litter sampling points were chosen along the diagonal line of the plot, including upper, middle, and lower sections, with each litter sampling point measuring 50 cm × 50 cm. And the litter sampling points were located outside the runoff plots of the rainfall experiments. The thickness of the non-decomposed and semi-decomposed litter layers in each quadrat was measured using a calibrated scale. Subsequently, the litter within the quadrat was systematically collected in layers, transported to the laboratory, and weighed.

Areas near the experimental site were selected as sampling points for determination of soil properties, such as soil water content, bulk density, total porosity, and saturated hydraulic conductivity (Fig. 1). The soil gravity water content was determined after oven drying at 105°C, for 24 h. Before the simulated rainfall in each plot, aluminum boxes of soil samples were collected at 10 cm intervals within the 0–1 m soil layer, with two replicates sampled per layer, and 20 soil samples were

collected from each plot. Following the simulated rainfall in each plot, the profile was simultaneously divided into 10 cm × 10 cm grids and 100 soil samples were collected to measure the soil moisture content after rain. A total of 2400 soil samples were collected. Changes in soil moisture content were obtained by subtracting the moisture content before rainfall from the moisture content after rainfall. The cutting ring method (Wang et al., 2008) was employed to measure the soil bulk density and total porosity. At each sampling point, ring-knife (diameter of 5 cm, length of 5 cm) soil samples were collected at 10 cm intervals within the 0–1 m soil layer. A total of 200 soil samples were collected. The soil samples were transported to the laboratory for drying to calculate bulk density and porosity. The constant head method (Klute and Dirksen, 1986) was used to measure the saturated hydraulic conductivity. The sampling method and quantity were same as the cutting ring method.

Following the simulated rainfall in each plot, root sampling points were identified within the 50 cm profile at the plot center. The focus of the root structure analysis was primarily on the biomass of roots with different diameters. The roots were further categorized into three diameter (*D*) classes: $D < 2$ mm, $2 \text{ mm} < D < 5$ mm, and $D > 5$ mm. The root soil samples were collected at 10 cm intervals in the 0–1 m soil layer to assess the root structure. Two replicate samples were collected per layer. A total of 320 root samples were collected. Subsequently, the root samples were transported to the laboratory, sieved, and thoroughly washed. After classification by diameter, all roots were dried (at 70°C, for 48 h) and weighed. In this study, the analysis of the roots of *R. pseudoacacia* was mainly focused on living roots.

2.3. Data analysis

2.3.1. Contribution of litter and roots to infiltration

To differentiate the impacts of litter cover and roots on infiltration, infiltration was quantified under three distinct conditions: *R. pseudoacacia* plantation with litter, *R. pseudoacacia* plantation without litter, and fallow slope land. Under identical simulated rainfall, we interpreted the disparity in infiltration between the *R. pseudoacacia* plantation with litter and fallow slope land as the outcome of the combined influence of litter and root factors. The difference in infiltration between the *R. pseudoacacia* plantation with and without litter was attributed to litter factors, whereas the difference between the *R. pseudoacacia* plantation without litter and fallow slope land was attributed to root factors. Infiltration volume of each plot was obtained by multiplying the above-mentioned soil moisture content change rate, bulk density and volume. The formulas used to calculate the infiltration volumes and contribution rates attributable to leaf litter and roots were as follows (Eqs. (1)–(5)):

$$\Delta I_{L+R} = \Delta I_L + \Delta I_R = I_{RP} - I_{FL} \quad (1)$$

$$\Delta I_L = I_{RP} - I_{RP-NL} \quad (2)$$

$$\Delta I_R = I_{RP-NL} - I_{FL} \quad (3)$$

where I_{RP} is the infiltration volume in the *R. pseudoacacia* plantation with litter (mm); I_{RP-NL} is the infiltration volume in the *R. pseudoacacia* plantation without litter (mm); and I_{FL} is the infiltration volume in the fallow slope land.

$$CR_L = \frac{\Delta I_L}{\Delta I_L + R} \quad (4)$$

$$CR_R = \frac{\Delta I_R}{\Delta I_L + R} \quad (5)$$

where CR_L is the contribution rate of litter (%) and CR_R is the contribution rate of roots (%).

2.3.2. Splitting of uniform and preferential flow

The staining tracer method mainly analyzes the infiltration and preferential flow distribution of the sample plot through the processing of the stained section image (Allaire et al., 2009). The processing of the stained image involved several steps, including geometric correction, white balance correction, color correction, and parameter calculation. Adobe Photoshop CS software (Adobe Systems Software Ireland Ltd., Dublin, Ireland) was used for geometric correction of the stained images, adjusting parameters such as image size, brightness, luminance, and threshold. This process rendered the stained areas black, whereas the unstained areas appeared white, and the images were then converted into binary bitmap format. MATLAB software (The MathWorks, Inc., Natick, MA, USA) was employed to process the binary bitmaps, establish matrices and assign values. Graphs depicting the staining proportion with soil layer depth were generated.

In this study, the area with a staining ratio of less than 80% was defined as the preferential flow area, and the other areas were defined as the uniform flow area (Mei et al., 2018). To exclude the impact of the boundary area of runoff plots on infiltration, the middle 60% area of the profile was selected as the area for analysis of infiltration, uniform flow, and preferential flow. The preferential infiltration volume (PIV) and its contribution to the total infiltration volume (TIV) were calculated using Eqs. (6) and (8), respectively (Van Schaik, 2009):

$$PIV = \sum_{i=0}^{i-ID} V_i - \sum_{j=0}^{j-UniFr} V_j \quad (6)$$

where ID is the infiltration depth (cm); $UniFr$ is the uniform infiltration depth (cm), that is, the infiltration depth of the uniform flow in the staining area (hereinafter referred to as the depth of uniform flow), which was calculated according to the infiltration depth of the staining area with a cover rate equal to 80%; and V_i and V_j represent the infiltration volume (mm) of the i th and j th soil layers.

$$V = 10 \times \sum_{u=1}^{u=6} (\theta_{tu} - \theta_{iu}) \times d \times V_s / 60 \quad (7)$$

where θ_{tu} is the soil moisture content after rainfall (%); θ_{iu} is the initial soil moisture content (%); u is the width of the soil sample sequence; d is the bulk density of the soil layer (g/cm^3); and V_s is the sample soil volume (cm^3).

$$C_{PF} = \frac{PIV}{TIV} \times 100 \quad (8)$$

where C_{PF} is the contribution of preferential flow to total infiltration (%).

2.3.3. Analytical method

Differences in infiltration and preferential flow infiltration between different stand ages were examined using One-way analysis of variance (ANOVA) and the least significance difference (LSD) post hoc test with SPSS 22.0 software (SPSS for Windows Release 22.0, SPSS Inc., Version

X; IBM, Armonk, NY, USA). The infiltration of three consecutive rainfall events in each experimental plot was considered separately as replicates in the calculations. And the significance level for mean difference was set at 0.05. An inter-group paired t -test was conducted and analyzed on the infiltration volume and the preferential infiltration contribution under different setting conditions. A Pearson correlation analysis was used to express the correlation between the infiltration rates and root parameters. And the threshold for statistical significance was $P < 0.05$.

3. Results

3.1. Underlying surface evolution

3.1.1. Litter

With increase in the age of *R. pseudoacacia* plantations, the litter density showed a trend to increase by approximately $40 \text{ g}/\text{m}^2$ per year. Within a specific range, the thickness of the non-decomposed and semi-decomposed litter layers increased with increase in plantation age. Beyond the 30-year age, the thickness of the non-decomposed layer decreased, and the proportion of the semi-decomposed layer was increased in the 40-year *R. pseudoacacia* plantation (Fig. 2).

To restore the original litter distribution in each plot as much as possible, the litter distribution in the test plots was supplemented and corrected according to the litter survey described above. The litter contents of the 18-, 25-, 30-, and 40-year *R. pseudoacacia* plantations were 600, 900, 1200, and $1200 \text{ g}/\text{m}^2$, respectively. The volume ratio of the non-decomposed and semi-decomposed layer of the 18-, 25-, and 30-year *R. pseudoacacia* plantations was 3:2, whereas that of the 40-year *R. pseudoacacia* plantation was 2:3.

3.1.2. Roots

Almost no roots were observed in the fallow slope land. Roots were abundant in the *R. pseudoacacia* plantations at each age, and the root content of the soil increased with increase in plantation age. However, the rate of increase decreased in the oldest *R. pseudoacacia* plantations (Fig. 3).

The roots in the *R. pseudoacacia* plantations with $D < 2 \text{ mm}$ were mostly distributed in the 0–60 cm soil layer and the root biomass decreased declined with increase in soil depth. The average root biomass per unit soil in the 0–1 m layer in the *R. pseudoacacia* plantations was ranked in the order 25 years > 30 years > 40 years > 18 years, and increased with increase in stand age but decreased after the stand exceeded a certain age (25–30 years). Roots with $2 \text{ mm} < D < 5 \text{ mm}$ diameter were mainly distributed in the 10–70 cm soil layer. With increase in soil depth, the biomass initially increased and thereafter

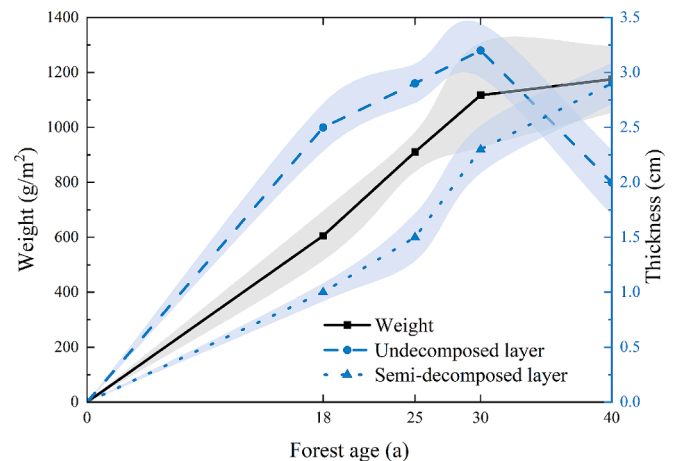


Fig. 2. Litter evolution characteristics at the study site on the Loess Plateau, China.

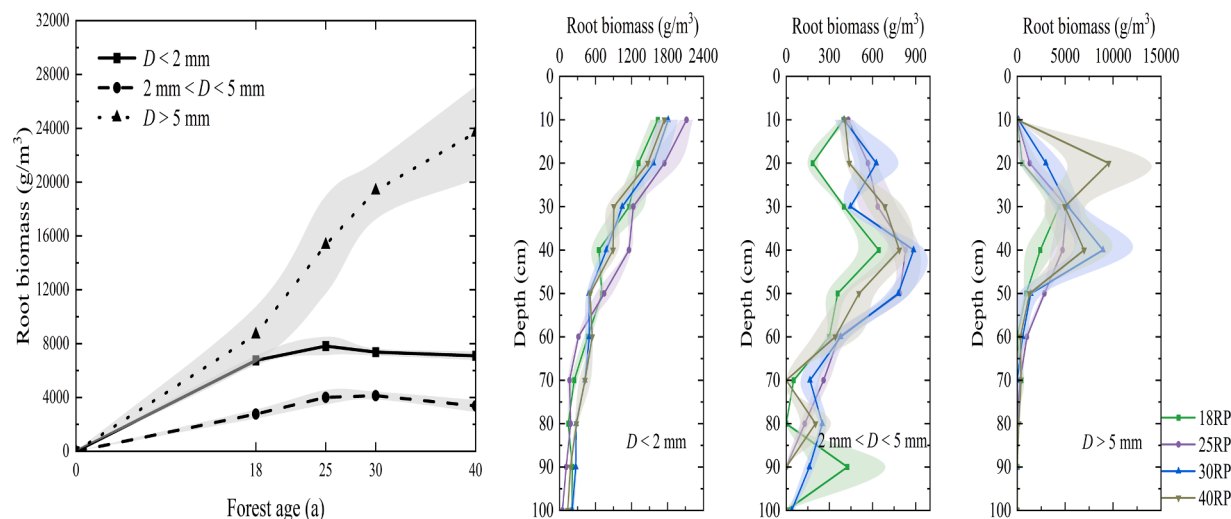


Fig. 3. Root system distribution characteristics at the study site. RP, *Robinia pseudoacacia* plantation with litter.

decreased, and the biomass in some soil layers fluctuated. These roots were mainly distributed in the 40 cm soil layer, but the overall root biomass in this diameter class was smaller than that of the other two diameter classes. The change in root biomass of this diameter class with stand age was identical to that of roots with $D < 2$ mm. Coarse roots in the diameter class $D > 5$ mm were mainly distributed in the 20–60 cm soil layer, and their biomass increased initially and thereafter decreased with increase in soil depth, with the largest distribution observed in the 20–40 cm soil layer. The biomass of coarse roots with $D > 5$ mm followed the rank order 40 years > 30 years > 25 years > 18 years. The root biomass increased significantly with the increase in stand age, and the difference between soil layers was significantly higher than that of the other two diameter classes. It was observed at the test site that a large proportion of the coarse roots with $D > 5$ mm in *R. pseudoacacia* plantations exceeded 10 mm in diameter, and their elongation direction was mainly horizontal.

3.1.3. Soil

The initial soil moisture content followed the order 40 years > fallow slope land > 30 years > 25 years > 18 years. The soil bulk density of *R. pseudoacacia* plantations showed a trend to increase in the 0–30 cm soil layer and tended to be stable below the 30 cm soil layer. With growth of the *R. pseudoacacia* plantations, the average bulk density

within the 1 m soil layer initially decreased and thereafter increased, and the total porosity changed in the opposite direction. The saturated hydraulic conductivity was greater in the 0–30 cm soil layer and gradually decreased and became stable below the 30 cm soil layer. The saturated hydraulic conductivity initially increased and thereafter decreased with plantation age (Fig. 4).

Comparing the changes in the three aforementioned parameters of the *R. pseudoacacia* plantations with soil depth, clear boundaries are observed in the 20–30 cm soil layer, and the changes tended to be stable below the 20–30 cm soil layer. However, no obvious boundary was detected in the fallow slope land. The surface soil on that slope was slightly loose, but did not differ notably from the soil at deeper layers.

3.2. Contribution of litter and roots to infiltration

In *R. pseudoacacia* plantations with litter cover, the ranking of the stands based on infiltration volumes was as follows: 30 years (216.00 mm) > 25 years (183.31 mm) > 18 years (174.75 mm) > 40 years (128.94 mm). In *R. pseudoacacia* plantations without litter cover, the ranking based on infiltration volumes was as follows: 25 years (168.28 mm) > 18 year (96.53 mm) > 30 years (76.30 mm) > 40 years (51.93 mm) (Table 1). Under the same simulated rainfall, the infiltration volumes in the *R. pseudoacacia* plantations increased with stand age.

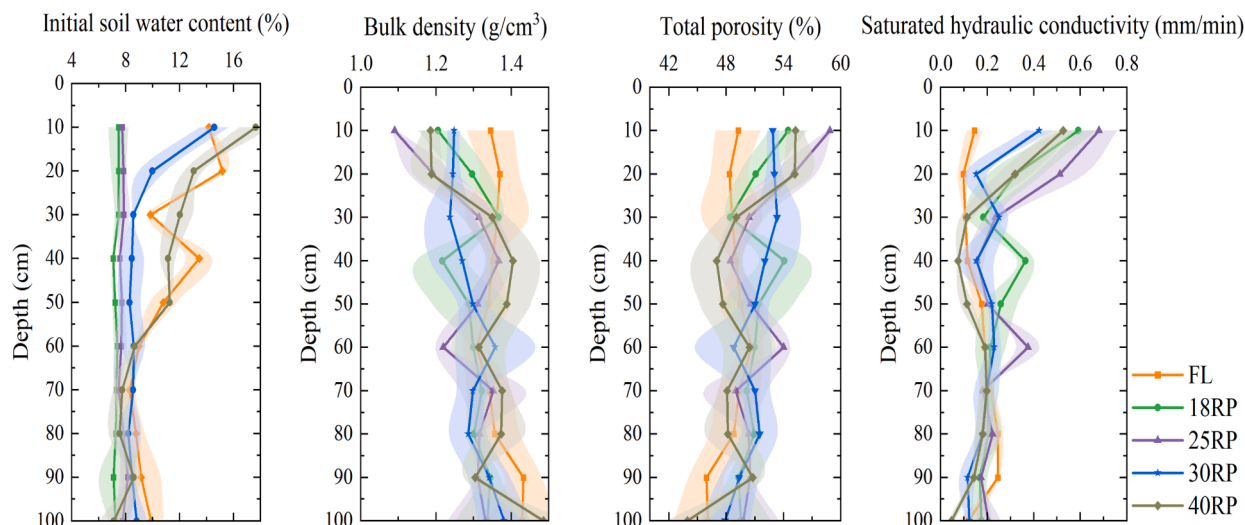


Fig. 4. Soil system distribution characteristics at the study site. RP, *Robinia pseudoacacia* plantation with litter; FL, fallow slope land.

Table 1

Total infiltration and preferential flow variables at the sampling sites.

Experiment	Sample plot	TIV/mm	PIV/mm	C _{PF} /%	Sample plot	TIV/mm	PIV/mm	C _{PF} /%
Rainfall I	18RP	58.52	—	—	18RP-NL	40.89	—	—
Rainfall II		58.26	—	—		25.42	—	—
Rainfall III		57.97	—	—		30.23	—	—
Total		174.75b	9.07b	5.19		96.53b	6.38bc	6.61
Rainfall I	25RP	61.24	—	—	25RP-NL	61.88	—	—
Rainfall II		61.12	—	—		47.99	—	—
Rainfall III		60.95	—	—		58.41	—	—
Total		183.31b	6.89b	3.76		168.28a	14.76a	8.77
Rainfall I	30RP	72.00	—	—	30RP-NL	28.13	—	—
Rainfall II		72.00	—	—		20.05	—	—
Rainfall III		72.00	—	—		28.13	—	—
Total		216.00a	16.87a	7.81		76.30bc	7.24b	9.49
Rainfall I	40RP	42.95	—	—	40RP-NL	22.01	—	—
Rainfall II		43.00	—	—		11.08	—	—
Rainfall III		42.98	—	—		18.84	—	—
Total		128.94c	6.80b	5.27		51.93c	4.27c	8.21
Rainfall I	FL	16.29	—	—	—	—	—	—
Rainfall II		14.24	—	—	—	—	—	—
Rainfall III		13.50	—	—	—	—	—	—
Total		44.03d	0.55c	1.24	—	—	—	—

NOTE: TIV, total infiltration volume; PIV, preferential infiltration volume; C_{PF}, contribution of preferential flow to total infiltration; RP, *Robinia pseudoacacia* plantation; RP-NL, *R. pseudoacacia* plantation without litter; and FL is the fallow slope land. Lowercase letters after the mean values indicate significant differences among different stand ages at the level of $P < 0.05$.

However, the infiltration volumes showed a trend to decrease after the stand age reached approximately 25–30 years. Under the same simulated rainfall, the infiltration volumes in the plantations were 2.93–4.91 times that in the fallow slope land. In the presence of litter, the average increase was 131.72 mm compared with that of the fallow slope land, and in the absence of litter, the average increase was 54.23 mm.

The paired t -test results indicated the infiltration volumes in *R. pseudoacacia* plantations with litter differed significantly from those of plantations without litter and the fallow slope land, whereas the infiltration volumes in *R. pseudoacacia* plantations without litter did not differ significantly from the fallow slope land. The contribution rates of preferential flow to total infiltration in these three plot types differed significantly (Fig. 5).

The changes in infiltration caused by the effects of litter cover showed a fluctuating trend with forest growth. The changes in infiltration caused by the root system initially increased and thereafter decreased (Fig. 6). We further calculated the contribution rates of litter and roots at the different stand ages. With increase in stand age from 18 to 40 years, litter contributed 60%, 11%, 82% and 91% to the increase in infiltration, and roots contributed 40%, 89%, 18% and 9%, respectively, to aforementioned four stands. For the four *R. pseudoacacia* plantations of different ages, the average litter contribution rate was 61%, the

average root contribution rate was 39%, and the contribution of litter was 1.56 times that of roots (Fig. 7). Overall, as the forest matured, there was an escalating trajectory observed in the contribution rate of litter to infiltration, concomitant with a diminishing trend in the contribution rate of roots to infiltration.

We regarded the contribution rate of litter and roots during simulated rainfall event I as the situation under low soil moisture content; correspondingly, the average values under simulated rainfall events II and III were regarded as the contributions under conditions of relatively high soil moisture content. With the change in soil moisture content from low to high, the contribution rate of litter to infiltration increased, and the contribution rate of roots to infiltration decreased in the four age groups of *R. pseudoacacia* plantations (Fig. 8).

3.3. The relationship of litter, roots, and infiltration

At 24 h after simulated rainfall, the soil moisture content in the upper soil layer of the four differently aged stands without litter was not significantly different in the vertical direction. However, the soil moisture content differed significantly in certain soil layers, namely, in the 40 cm, 60 cm, 30 cm, and 15 cm soil layers in the 18-, 25-, 30-, and 40-year *R. pseudoacacia* plantations, respectively (Fig. 9). Soil moisture was mainly supplied to the 50–100 cm layers. In the presence of litter, the soil moisture content differed significantly changes in the 80–90 cm layers in the 18-, 25-, 30-, and 40-year *R. pseudoacacia* plantations (Fig. 9). The soil moisture was recharged to the 110–130 cm layers, which was approximately 40 cm deeper than in the stands without litter.

Infiltration in the plots was dominated by uniform infiltration and supplemented by preferential infiltration, but the preferential infiltration depths differed notably among the *R. pseudoacacia* plantations with different litter cover and different stand ages, and in fallow slope land. The Y value corresponding to $X = 0$ represented the maximum staining depth of the profile, and the Y value corresponding to $X = 80\%$ represented the depth of uniform flow in the stained area of the profile. In the absence of litter cover, the 18-year stand had a uniform staining distribution in the 0–20 cm soil layer, branches with different staining depths occurred in the 20–31 cm soil layer, and the thickness of the preferential flow was 11 cm. The 25-year stand had a uniform staining distribution in the 0–34 cm soil layer, and branches with different staining depths occurred in the 33–54 cm soil layer. The thickness of the preferential

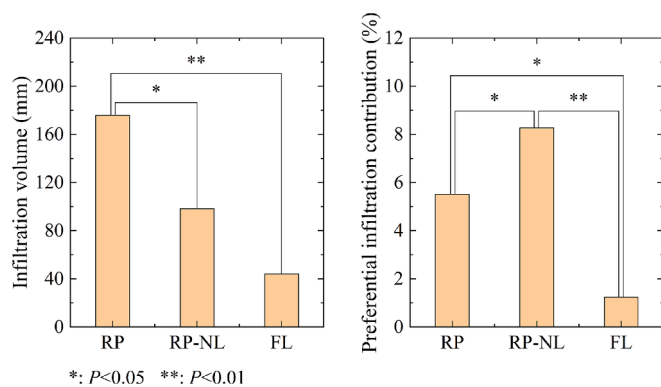


Fig. 5. Results of paired t -test analysis of infiltration volume and preferential infiltration contribution. RP, *Robinia pseudoacacia* plantation with litter; RP-NL, *R. pseudoacacia* plantation without litter; FL, fallow slope land.

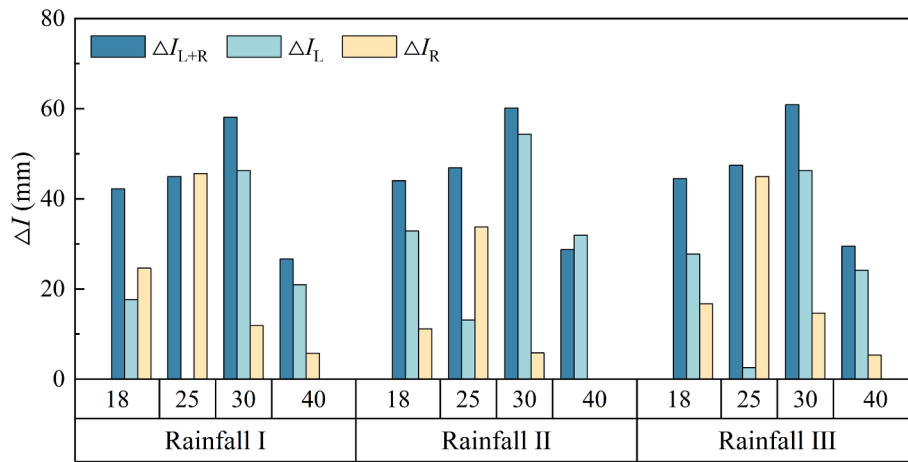


Fig. 6. Differences in infiltration volume under different experimental conditions. ΔI_{L+R} , difference in infiltration between the *Robinia pseudoacacia* plantation with litter and fallow slope land; ΔI_L , difference in infiltration between the *R. pseudoacacia* plantations with and without litter; and ΔI_R , difference in infiltration between the *R. pseudoacacia* plantation without litter and fallow slope land; 18, 18-year *R. pseudoacacia* plantations; 25, 25-year *R. pseudoacacia* plantations; 30, 30-year *R. pseudoacacia* plantations; 40, 40-year *R. pseudoacacia* plantations.

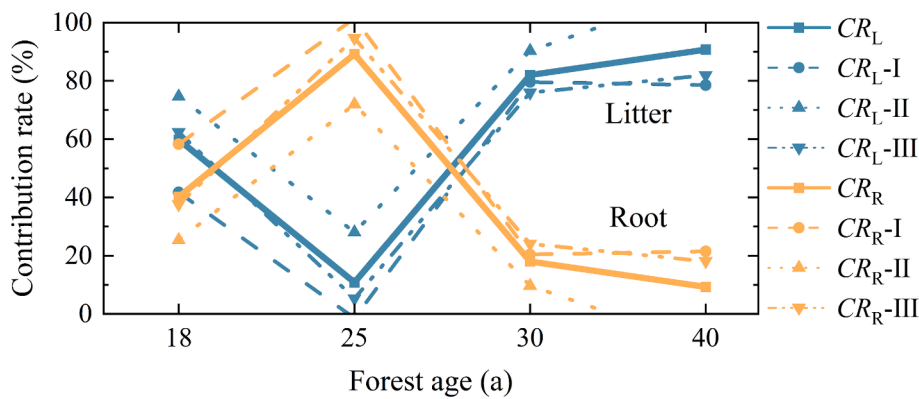


Fig. 7. Contribution rates of litter and roots to changes in infiltration. CR_L , contribution rate of litter; CR_R , contribution rate of roots; I, rainfall event I; II, rainfall event II; and III, rainfall event III.

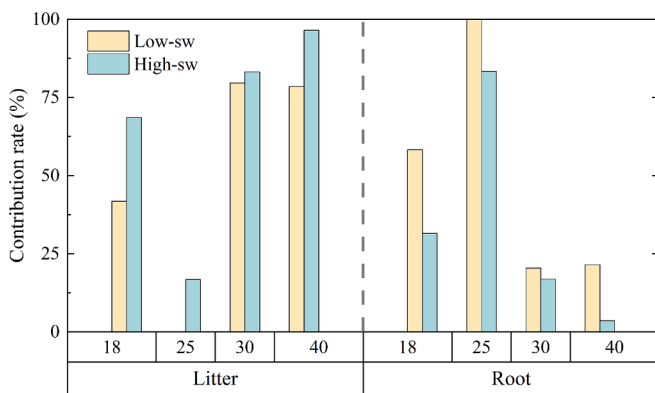


Fig. 8. Changes in contribution rate to infiltration under different soil moisture content. Low-sw, low soil moisture content; High-sw, high soil moisture content; 18, 18-year *Robinia pseudoacacia* plantations; 25, 25-year *R. pseudoacacia* plantations; 30, 30-year *R. pseudoacacia* plantations; 40, 40-year *R. pseudoacacia* plantations.

flow was 11 cm. The 30-year stand had a uniform staining distribution in the 0–9 cm soil layer, different staining branches occurred in the 9–22 cm soil layer, and the thickness of the preferential flow was 13 cm. The 40-year stand had a uniform staining distribution in the 0–5 cm soil

layer, different stained branches were detected in the 5–13 cm soil layer, and the thickness of the preferential flow layer was 8 cm (Fig. 10). In the presence of litter cover, the staining depth of preferential flow in the four differently aged *R. pseudoacacia* plantations increased greatly, the maximum staining depth from deep to shallow layers was 58 cm (25-year), 57 cm (30-year), 55 cm (18-year), and 50 cm (40-year) (Fig. 10). Compared with the case without litter cover, the maximum staining depth of *R. pseudoacacia* plantations at 1, 25, 30, and 40 years increased by 24, 4, 35, and 37 cm, respectively, and the thickness of preferential flow increased by 1, 1, 2, and 14 cm, respectively. Litter cover had little effect on the thickness of preferential flow. Moreover, the C_{PF} with litter was, on average, 67% of the C_{PF} without litter (Table 1).

The fallow slope land showed relatively uniform staining in the 0–14 cm soil layer. Comparing the stained lower edge lines in the section staining map, it was observed that the lower edge line for fallow slope land was much smoother. Multiple peaks were detected in the stained lower edge lines of *R. pseudoacacia* plantations. In the absence of a root system (i.e., fallow slope land), the infiltration depth was essentially unchanged and relatively uniform, and the contribution of preferential infiltration to total infiltration was significantly reduced (Table 1).

Significant differences were observed in the effects of the different diameter classes of living roots on the infiltration process in *R. pseudoacacia* plantations. Fine roots with $D < 2$ mm were significantly positively correlated with the total infiltration coefficient, uniform infiltration coefficient and preferential infiltration coefficient. Roots

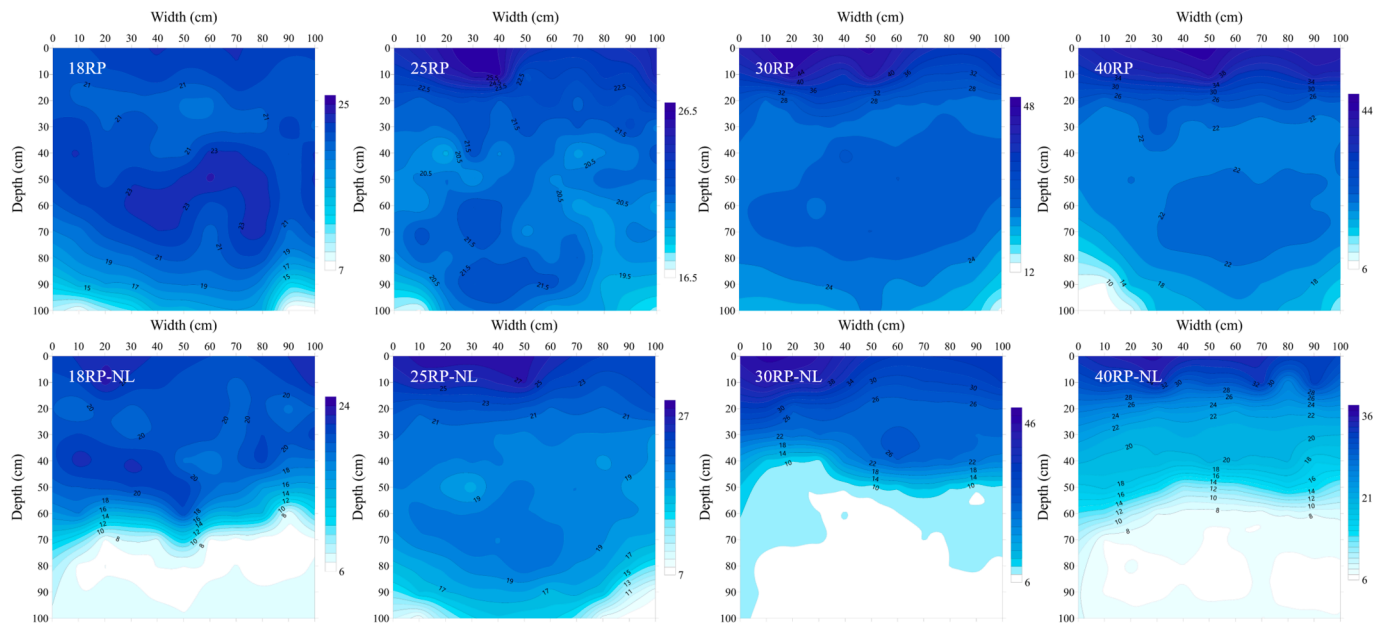


Fig. 9. Moisture distribution in the soil profile 24 h after simulated rainfall. RP, *Robinia pseudoacacia* plantation with litter; RP-NL, *R. pseudoacacia* plantation without litter.

with $2 \text{ mm} < D < 5 \text{ mm}$ had a significant positive correlation with the total infiltration coefficient and uniform infiltration coefficient. Fine roots had a significant positive effect on the infiltration capacity of *R. pseudoacacia* plantations, especially fine roots with $D < 2 \text{ mm}$, which had a significant effect in promoting preferential flow. Roots with $D > 5 \text{ mm}$ were negatively correlated with the total infiltration coefficient and uniform infiltration coefficient, which might have an inhibitory effect on uniform infiltration (Table 2).

4. Discussion

4.1. Litter cover and infiltration

Litter traits for plantation species, including litter density, and degree of decomposition, play a crucial role in the interception, infiltration, and redistribution of precipitation (Zhou et al., 2015b; Oliveira et al., 2019). After *R. pseudoacacia* plantations had entered the mature and aging stages, the density and decomposition of litter increased (Fig. 2). The contribution rate of litter to infiltration showed an increasing trend (Fig. 7). However, the infiltration volume and the increment in infiltration influenced by litter in the *R. pseudoacacia* plantations did not show a continuous increase, with both variables exhibiting low values during the aging stage (Fig. 6). The reason for this might be that in the 30-year *R. pseudoacacia* plantation, the litter density 1200 g/m^2 had already reached its maximum effect on infiltration. In the 40-year *R. pseudoacacia* plantation, although the proportion of semi-decomposed litter layer increased, its enhancing effect on infiltration ceased to grow. At the 40-year plantation site, the soil bulk density of the soil layer below 20 cm in the *R. pseudoacacia* plantations increased significantly, and the saturated hydraulic conductivity decreased (Fig. 4). Increases in litter density and degree of decomposition failed to sufficiently enhance the surface soil structure to offset deficiencies in the deeper soil structure (Brantley et al., 2019; Zhao et al., 2019). Cui and Pan (2023) conducted field in-situ experiments in a *Pinus tabulaeformis* plantation and showed that the positive effect of litter on infiltration ceases to be intensified after the litter density surpasses 1200 g/m^2 . In both the in-situ experiments conducted in *R. pseudoacacia* plantation and *Pinus tabulaeformis* plantation, around 1200 g/m^2 might represent the threshold density of litter for infiltration influence, which could be applicable in certain forested areas. This value might also vary owing to

a combination of factors such as leaf shape, decomposition level, and soil conditions (Cui et al., 2022; Zhang et al., 2020; Stankova and Aranda, 2017).

At the 25-year plantation site, the contribution of litter to infiltration was lower than that at the other study sites. Conversely, the contribution of roots to infiltration was relatively higher. The reason for this might be that, in the 25-year *R. pseudoacacia* plantation, the surface soil bulk density was 8%–13% lower than that of others, and the surface soil saturated hydraulic conductivity reached 0.68 mm/min (Fig. 4), indicating a stronger infiltration capacity compared to other sites. Under the simulated rainfall in the present experiment, substantial infiltration was achieved even in the absence of litter cover. Whereas with litter cover, the maximum infiltration capacity might not be reached, suggesting the potential for further increases in infiltration with higher rainfall intensities. Consequently, the contribution of litter to infiltration was underestimated (Su et al., 2023).

4.2. Roots and infiltration

The growth extension, and alternate production and death, of roots may cause a change in soil structure and then change the rainfall infiltration process. In the current study, the *R. pseudoacacia* plantations grew vigorously and rapidly during the juvenile and mature stages, and the root biomass in each diameter class increased rapidly. In particular, the fine-root biomass reached a maximum. The biomass and proportion of coarse roots increased obviously during the aging stage (Fig. 3). Claus and George (2005) showed that in *Fagus sylvatica* L., *Picea abies* (L) Karst. and *Quercus cerris* L. plantation, fine-root biomass initially increased and thereafter decreased with growth of the forest, and the highest fine-root biomass was recorded in adult but not mature stands. John et al. (2001) showed that annual fine-root production showed a marked decrease but coarse root production increased, with forest growth. For different tree species, variation exists in the critical age at each growth stage. However, the overall evolution pattern of root systems with forest age in most plantations is similar to that observed in *R. pseudoacacia* plantations. Therefore, the research results for *R. pseudoacacia* plantations might also be applicable to for other plantations.

The present study also observed differences in the effects of roots of different diameters on infiltration in *R. pseudoacacia* plantations. Roots with $D < 5 \text{ mm}$ had a significant effect in promoting the infiltration

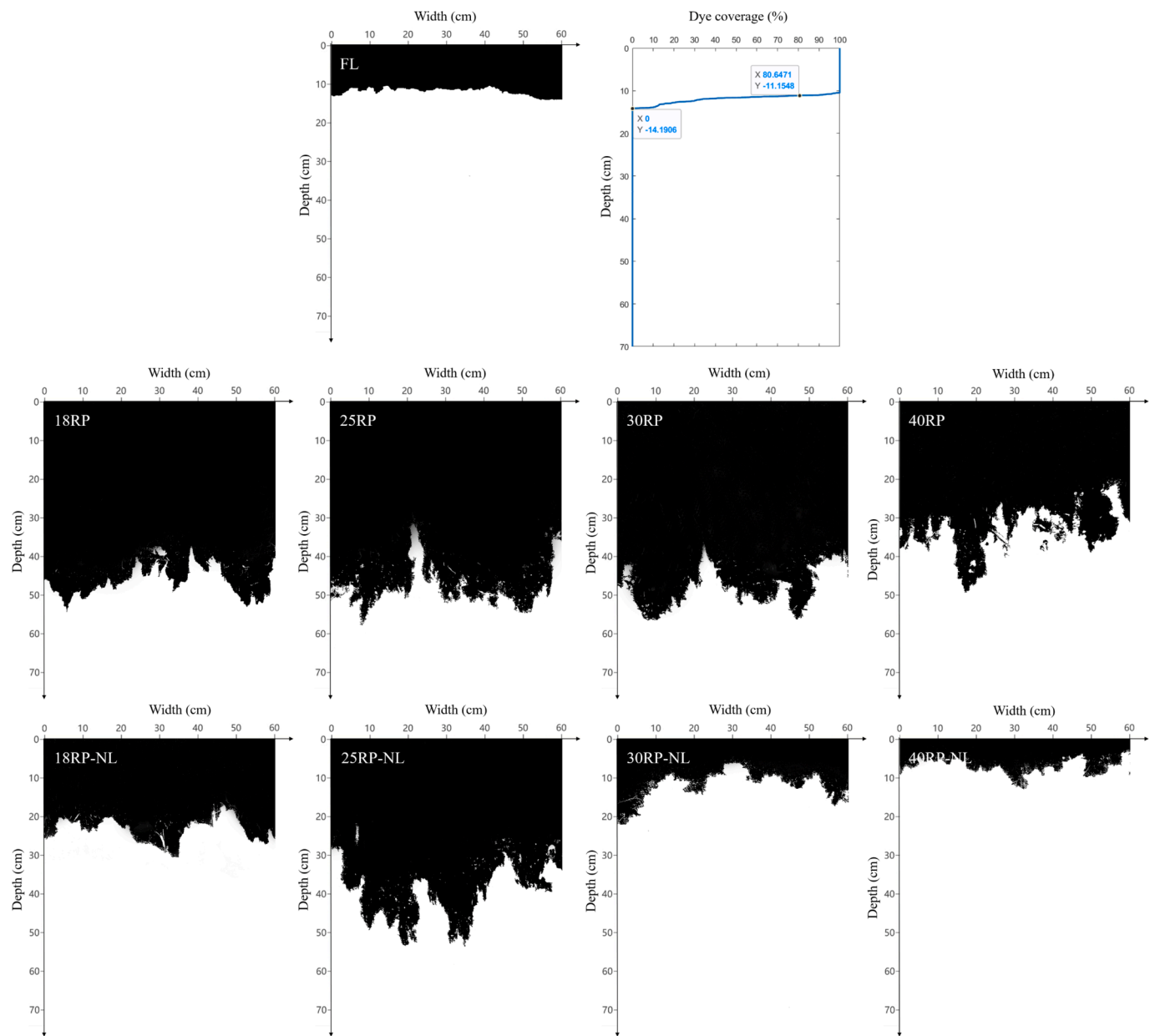


Fig. 10. Section staining diagram. FL, fallow slope land; RP, *R. pseudoacacia* plantation; RP-NL, *R. pseudoacacia* plantation without litter; Dye coverage, proportion of stained area in each soil layer.

Table 2
Correlation coefficients between infiltration characteristics and root parameters.

Infiltration characteristic	Root biomass		
	$D < 2$	$2 < D < 5$	$D > 5$
Infiltration volume	0.669*	0.527	-0.483
Uniform infiltration volume	0.650**	0.514**	-0.502*
Preferential infiltration volume	0.831**	0.528**	-0.278*

NOTE: D (mm) = Root diameter ; *: $P < 0.05$; **: $P < 0.01$.

capacity of *R. pseudoacacia* plantations, especially fine roots, which had a significant positive effect on preferential flow, whereas coarse roots may have an inhibitory effect on uniform infiltration (Table 2). Fine roots are the most active and sensitive diameter class of plant roots, and plants mainly absorb water and nutrients in the soil through fine roots (Aun et al., 2021; Wambsganss et al., 2021; Matamala et al., 2003). Fine roots were mainly distributed in the 0–40 cm soil layer (Fig. 3), and were

interlaced to form large spatial networks and root channels, which loosened the originally dense soil, decreased the bulk density, and increased the porosity, and thus the soil saturated hydraulic conductivity was further increased (Silva et al., 2021). Thus, preferential flow and the downward migration of water are promoted (Chen et al., 2022). Nevertheless, coarse roots might block the original soil pores, and most coarse roots were horizontally elongated in *R. pseudoacacia* plantations, which would disrupt the longitudinal connectivity of the pores and hinder water infiltration (Luo et al., 2019).

In *R. pseudoacacia* plantations, the roots contribution to infiltration was higher during the juvenile and mature stages, whereas a decreasing trend was observed during the aging stage (Fig. 7). The reason for this might be that in the early and middle stages of *R. pseudoacacia* plantations, the proportion of fine roots was relatively high (Fig. 3), leading to a 4% average increase in soil porosity at 0–1 m depth (Fig. 4), thereby enhancing infiltration. As the *R. pseudoacacia* plantations entered the maturity and aging stage, the proportion of thick roots increased, the

root system might change from a predominantly longitudinal elongation to mainly transverse thickening, soil was compacted by squeezing (Yuan and Chen, 2010; Shively et al., 2022; Wasylw and Karst, 2020). The soil bulk density in layers affected by roots exhibited an increase of 4–9%, whereas the saturated hydraulic conductivity exhibited a decrease of 49%–63% (Fig. 4), resulting in a decline in infiltration. Consequently, it was further postulated that changes in soil bulk density exert a more pronounced influence on infiltration. Soil could also have been influenced by root secretions, which were capable of reducing infiltration resistance, enhancing microbial activity, thereby promoting the formation of preferential pathways, and increasing water infiltration in the soil (Oleghe et al., 2017; Zhang et al., 2020). Chen et al. (2023) showed that with increasing stand age, the exudation rate of root secretions decreased, resulting in a reduced facilitating effect of roots on infiltration during the aging stage. Previous research has suggested that root decay significantly increases the number of large soil pores, thereby promoting infiltration (Xiao et al., 2024). The present study found that decayed roots in *R. pseudoacacia* plantations were relatively rare and distributed deeper in the soil layers, resulting in minimal impact on infiltration.

4.3. Comprehensive effect

This study observed that the average contribution rate of litter and roots to infiltration were 61% and 39%, respectively. The positive effect of litter on promoting infiltration was 1.56 times that of roots, and litter weakened the effect of roots on infiltration (Fig. 7). The promotive effect of litter and roots on infiltration has been confirmed in previous studies, but the specific contribution rates of litter and roots to infiltration were not stated, nor were they quantitatively compared. Litter mainly promotes rainfall infiltration on hillslopes by retaining runoff, changing the surface soil properties through decomposition, and hindering the formation of surface crusts (Wang et al., 2020). Roots mainly alter the soil structure and enhance the saturated hydraulic conductivity (Vannoppen et al., 2017). *R. pseudoacacia* is a broad-leaved tree. In comparison with coniferous forests, litter in broad-leaved forests is more readily decomposed, has a more noticeable effect on soil improvement, and is less prone to lateral flow (Kim et al., 2014; Sidle et al., 2010; Stan et al., 2017). Furthermore, the simulated rainfall applied in the present study equated to a heavy rainfall event. Under heavy rainfall, the impact of litter on the surface crust is more significant. In the absence of litter, the fine particles of the surface soil are rearranged owing to the impact of raindrops, blocking the soil pores and forming surface crusts, thus significantly hindering rainfall infiltration, and further weakening the role of roots in promoting infiltration (Hu et al., 2012; Neave and Rayburg, 2007; Liu et al., 2019). Therefore, in the present study, the impact of litter on infiltration was significantly higher than that of roots. However, the effect of litter in coniferous plantations might not be as great (Rajão et al., 2023).

The majority of the effect of litter on infiltration in *R. pseudoacacia* plantations was on uniform infiltration, promoting its depth and proportion (Fig. 10). Previous studies on the impact of litter on infiltration did not distinguish between uniform and preferential infiltration. The reason may be that the litter was mainly evenly distributed on the soil surface and the spatial heterogeneity of its impact on infiltration was small (Sanders et al., 2012). By comparing the preferential infiltration characteristics of *R. pseudoacacia* plantations and fallow slope land, we determined that the root system mainly promoted total infiltration by promoting preferential infiltration (Fig. 10). This may be because processes such as root elongation, decay, and decomposition create pores (York et al., 2016; Geng et al., 2022), giving rise to preferential flow channels and thereby promoting preferential infiltration (Vannoppen et al., 2017). Zhang et al. (2015) reached the same conclusion. The main influencing factors of roots on soil and of soil on preferential flow are strongly associated with soil porosity (Erktan et al., 2016; Hudek et al., 2017). The water supply of *R. pseudoacacia* plantations was dominated

by uniform infiltration, supplemented by preferential infiltration, which might explain why litter had a more strongly significant impact on infiltration. In the future management of watersheds, consideration should be given to adjusting the structure of forest litter to regulate river runoff to achieve a reasonable distribution of water resources.

The present study conducted field in-situ experiments on plots with similar conditions such as slope and aspect, systematically analyzing the entire process of afforestation over decades. We quantitatively distinguished the contributions of litter and roots to infiltration, explained the infiltration mechanisms, and derived several findings, providing insights for the management of afforested areas. However, the study is subject to certain limitations. The experiment was established based on the method of “Using space instead of time,” and it was not possible to avoid differences in soil characteristics before vegetation restoration was initiated. Although it was generally believed that the soil in the loess area was relatively uniform, a few outliers were observed in the data. In subsequent studies, a long-term series of observations can be conducted in the same experimental area to reduce variables as much as possible. Additionally, needle-type simulated rainfall was employed for artificial rainfall simulation, which differed somewhat from actual rainfall, with potentially smaller raindrops and lacking a raindrop spectrum. The kinetic energy of raindrops might have been lower compared to actual rainfall, and the crust effect of bare soil without litter cover might not have been as pronounced as in reality. Consequently, the observed results might have amplified infiltration, underestimating the effect of litter.

5. Conclusions

The infiltration experiment was performed in field plots under in situ simulated rainfall. *Robinia pseudoacacia* plantations aged 18, 25, 30, and 40 years were selected to investigate the influence of litter cover and roots on infiltration processes on hillslopes in different stand ages. The plantations with and without litter cover were compared to separate the relative importance of an increase in infiltration.

The litter accumulation increased with stand growth. When *R. pseudoacacia* plantations exceeded 30 years of age, litter input decreased, and the proportion of the semi-decomposed layer increased. The root content of the soil increased with increase in plantation age, whereas fine-root biomass initially increased, but decreased after the stand exceeded 25 years of age. Litter and roots played important roles in the infiltration process and their relative importance changed with stand age. In *R. pseudoacacia* plantations with litter cover, the infiltration volumes were 128.94–216 mm; in *R. pseudoacacia* plantations without litter cover, the infiltration volumes were 51.93–168.28 mm. Under the same simulated rainfall, the infiltration volumes in the plantations initially increased and thereafter decreased with forest growth, which were 2.93–4.91 times that of the fallow slope land. The average contribution of litter to the increase in infiltration was 1.56 times greater than that of roots. The contribution of litter generally increased with stand age. Moreover, litter cover and the root system played crucial roles in strengthening uniform infiltration and preferential infiltration, respectively. Under the influence of litter, the contribution of preferential flow to total infiltration was 67% that of the condition without litter. Roots with $D < 5$ mm had a significant effect in promoting the infiltration capacity of *R. pseudoacacia* plantations, especially fine roots, whereas coarse roots may have an inhibitory effect on uniform infiltration. This research improves our understanding of the impact of plantations on watershed hydrological processes and provides a scientific basis for sustainable forest management.

CRedit authorship contribution statement

Fengjiao Niu: Writing – original draft, Investigation, Formal analysis, Conceptualization. Chengzhong Pan: Writing – original draft, Supervision, Conceptualization. Lan Ma: Writing – review & editing.

Yongsheng Cui: Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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